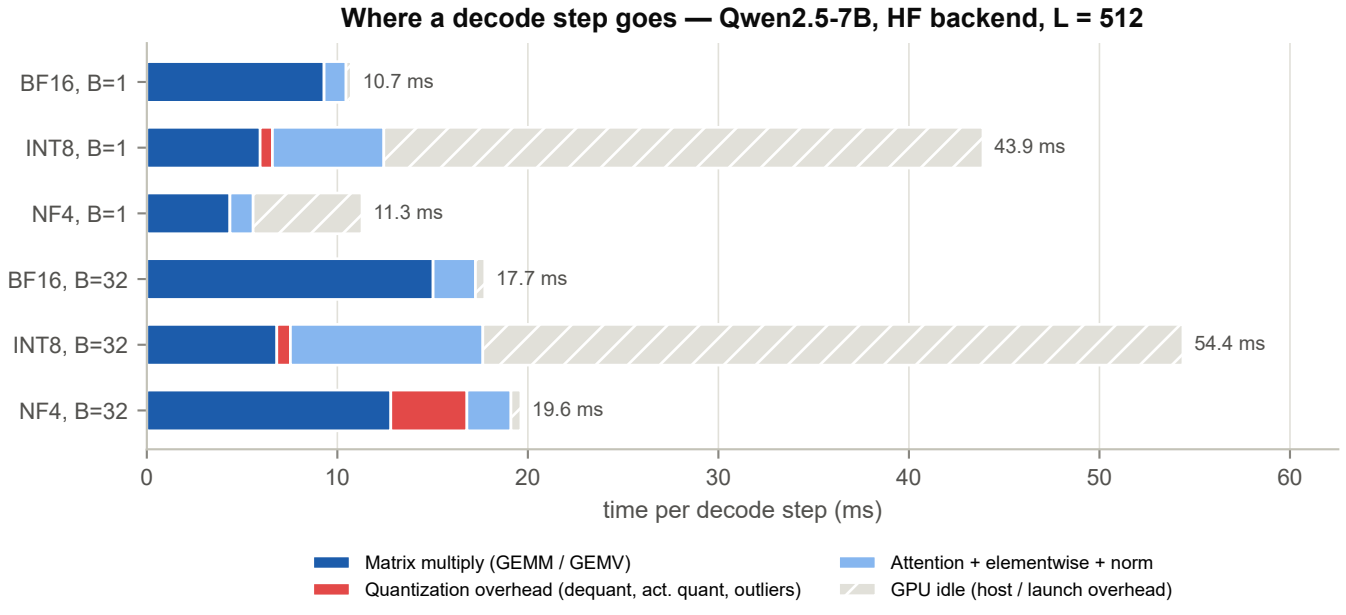
Where a decode step goes — Qwen2.5-7B, HF backend, L = 512
BF16, B=1
INT8, B=1
NF4, B=1
BF16, B=32
INT8, B=32
NF4, B=32
10.7 ms
43.9 ms
11.3 ms
17.7 ms
54.4 ms
19.6 ms
0
10
20
30
40
50
60
time per decode step (ms)
Matrix multiply (GEMM / GEMV)
Quantization overhead (dequant, act. quant, outliers)
Attention + elementwise + norm
GPU idle (host / launch overhead)